Rethinking Token Valuation for Digital Platforms

MSc in Quantitative Finance

Alma Mater Studiorum - Universitá di Bologna

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Rethinking Token Valuation

Platform ***tokens do not fit*** neatly into ***traditional asset-pricing*** categories.



CWL provides the **benchmark mechanism**: token prices can ***internalize*** expected platform ***growth*** and ***strengthen adoption incentives***.



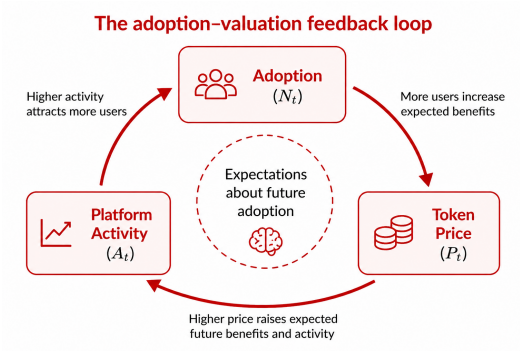
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RESEARCH QUESTION

*How should a **token** be **valued** when it is both a **financial asset** and part of the **platform**'s economic mechanism?*

The Framework

1 Rethinking Token Valuation



Model Ingredients

- Platform **activity** / productivity: A_t
- Continuum of heterogeneous **users**
- **Token** is the means of **payment** on the platform
- **Users decide** whether to adopt at time t

Core Mechanism

- Users **adopt if** expected **benefit** $\geq$ **cost**
- Token **price** P_t reflects expected **future adoption**
- Feedback **loop**

Equilibrium Pricing Equation (CWL)

The token price $P(A_t)$ is determined in equilibium by the **no-arbitrage condition** applied to its dynamics.

$$rP(A_t) = \underbrace{P'(A_t)A_t\mu^A}_{\text{Growth Term}} + \underbrace{\frac{1}{2}P''(A_t)(A_t\sigma^A)^2}_{\text{Risk (convexity) Term}} + \underbrace{(1-\alpha)\left(\frac{N(A_t)S(A_t)A_t}{P(A_t)M}\right)^{\alpha}P(A_t)}_{\text{Adoption Utility Term}}$$

Growth

Faster adoption growth **increases** expected returns and token **value**

Risk

Uncertainty increases the **convexity** value embedded in the token

Adoption

User **participation generates** the platform **utility** supporting value

Three Extensions Developed in the Thesis

1 Rethinking Token Valuation

∞

Infinite Horizon

The platform is assumed to exist forever

↓

Endogenous Adoption

Network effects drive user adoption and token demand

↓

Fixed Token Supply

Token supply is taken as given, issuance is not a policy choice

→

Finite maturity

How does platform duration affect token value?

External Competition

When does the token dominate an outside asset?

Issuance Design

How should supply policy balance growth and dilution?



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Why Horizon Matters

2 From Infinite to Finite Horizon

Real platforms have finite lifetimes

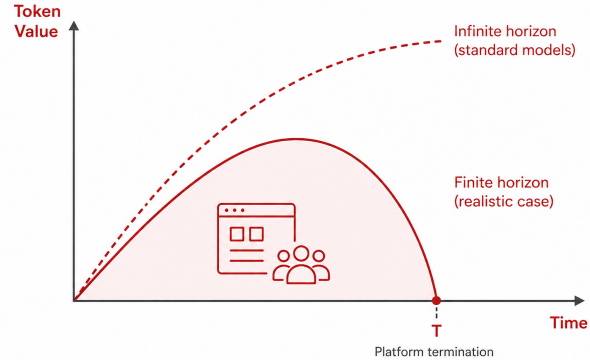
Platforms can fail, be abandoned, or lose relevance

Token value depends on future expectations

Value is driven by expected future adoption and usage

The horizon assumption affects valuation

The length and uncertainty of the horizon shape expectations and prices



Key takeaway: If the platform may end, expectations and token value must adjust. The horizon is not a detail: it is a driver value.

Types of termination

2 From Infinite to Finite Horizon

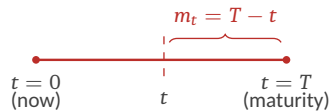
Deterministic Maturity

(Known Maturity T)

- A fixed terminal date $T > 0$ is given.
- We track the remaining time to maturity:

$$m_t = T - t$$

- State space for time: $m_t \in [0, T]$.



Access rights **deterministically expire** at $t = T$.
As $m_t \downarrow 0$, the remaining horizon *shrinks to zero*.

vs.

Stochastic Maturity

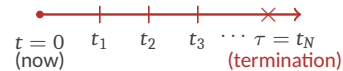
(Hazard Rate $\delta > 0$)

- Termination time τ is random.
- We assume $\tau \sim \text{Exponential}(\delta)$:

$$\mathbb{P}(\tau > t) = e^{-\delta t}, \qquad t \geq 0.$$

- Equivalently, termination follows a Poisson process:

$$N_t \sim \text{Poisson}(\delta t), \quad \tau = \inf\{t \geq 0 : N_t = 1\}.$$



Access **rights terminate** at the **random time** τ.
There is always a *positive probability of termination* in any interval.

Economic Intuition

2 From Infinite to Finite Horizon

WHAT CHANGES

Shorter Capitalization Horizon
Fewer future benefits can be capitalized today.

Expiration Risk
Random termination adds uncertainty and lowers value.

Valuation Drag
As maturity approaches, present value naturally declines.

WHAT STAYS THE SAME

Lower Effective Carry Cost
Tokens still let users capitalize expected growth.

Feedback Loop
More adoption increases value, which attracts more adoption.

Rational Expectations
Expiration is anticipated and priced ex ante.

The Mechanism Intact



BIG PICTURE: **Finite duration attenuates** token **values**, but tokenization **continues** to align incentives and **transform** expected future **growth into value** today.

Deterministic Finite-Maturity Equilibrium

2 From Infinite to Finite Horizon

User problem

Each **user chooses** the real **token balance** $x_{i,t}$ **to maximize** instantaneous **utility** net of the effective carry cost.

$$\max_{x_{i,t} \geq 0} \left[x_{i,t}^{1-\alpha} (N_t A_t e^{u_i})^{\alpha} - \phi - (r - \mu_t^P) x_{i,t} \right]$$

Expected token appreciation μ_t^P lowers the effective participation cost.

$\longrightarrow$

Market clearing

The token market links **price**, **adoption**, and **expected appreciation**.

$$P(A_t, m_t) = \frac{N_t S_t A_t}{M} \left(\frac{1-\alpha}{r - \mu^P(A_t, m_t)} \right)^{\frac{1}{\alpha}}$$

Market clearing transforms individual demand into an **equilibrium price**.

SUBSTITUTING market clearing into the asset pricing condition yields a finite-maturity PDE.

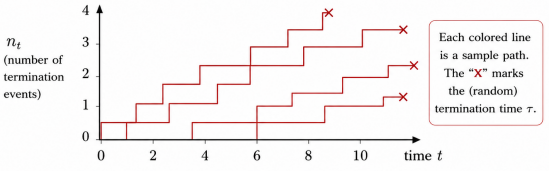
$$rP(A_t, m_t) = \mu^A A_t P_A + \frac{1}{2}(\sigma^A)^2 A_t^2 P_{AA} \boxed{-P_m} + (1-\alpha) \left(\frac{N(A_t, m_t) S(A_t, m_t) A_t}{P(A_t, m_t) M} \right)^{\alpha} P(A_t, m_t)$$

The remaining **maturity** $m_t = T - t$ becomes an **additional state variable**.

Stochastic Termination Equilibrium

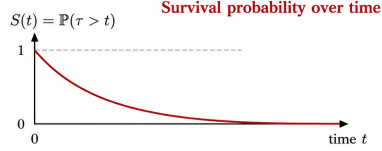
2 From Infinite to Finite Horizon

Poisson process: possible termination paths



Termination can occur at any instant; we do not know when.

Survival probability over time



$$S(t) = e^{-\delta t}$$

The probability that the platform is still active decays exponentially over time.

Equilibrium pricing equation

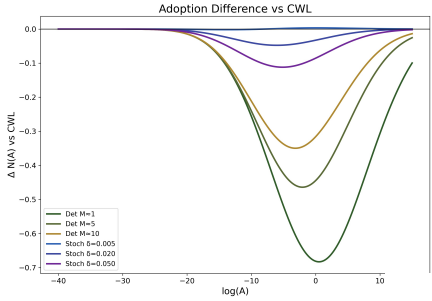
$$rP(A_t) = P_A A_t \mu^A + \frac{1}{2} P_{AA} A_t^2 (\sigma^A)^2 \quad \boxed{-\delta P(A_t)} + \\ + (1-\alpha) \left(\frac{N(A_t) S(A_t) A_t}{P(A_t) M} \right)^{\alpha} P(A_t).$$

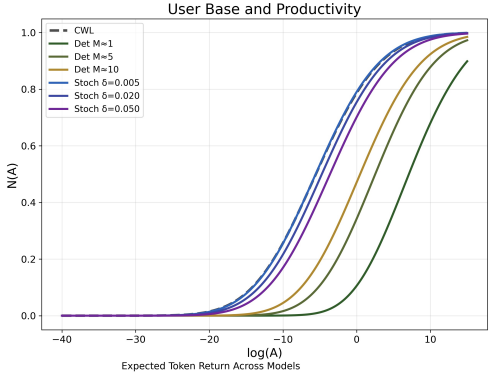
- A **constant hazard rate** δ **lowers** the **continuation value** of the token by adding a killing term.
- **No extra time-state variable** is required.

The term $\delta P(A_t)$ discounts continuation value by incorporating the probability of platform termination.

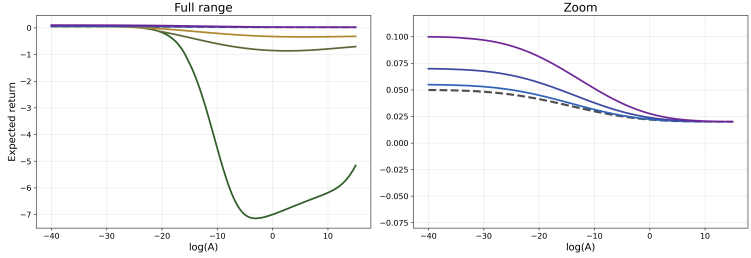
Numerical Results

2 From Infinite to Finite Horizon





- **Adoption** is **delayed** under finite horizons.
- **Deterministic** maturity creates **larger distortions** than stochastic termination.
- **Hazard-based** termination remains **close** to the **CWL** benchmark.
- **Valuation compression increases** as maturity shortens.





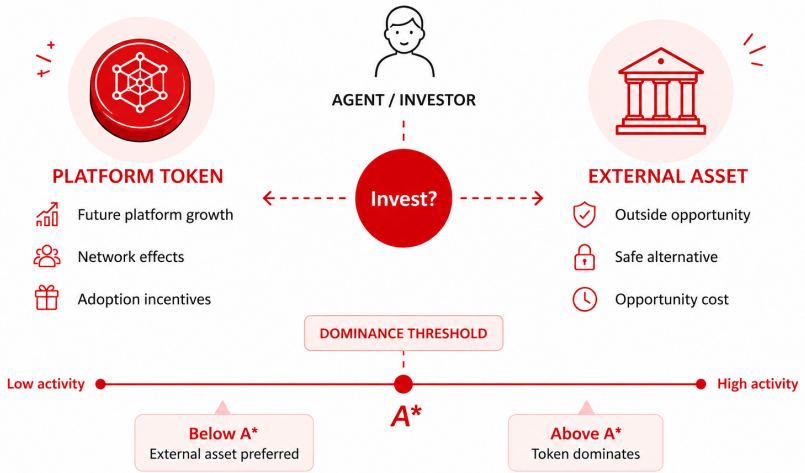
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From Valuation to Competitiveness

3 When Does the Token Dominate?



Token vs External Asset

3 When Does the Token Dominate?

Dominance function

$$D(A;X) = P(A) - X$$

$D(A;X) < 0$

External asset dominates

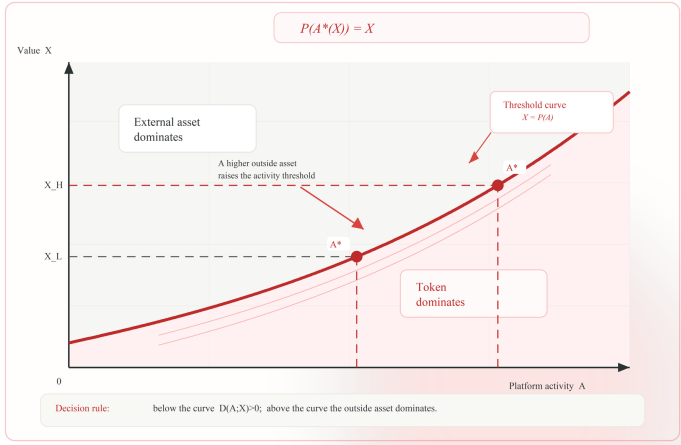
$D(A;X) \geq 0$

Token dominates

Threshold condition

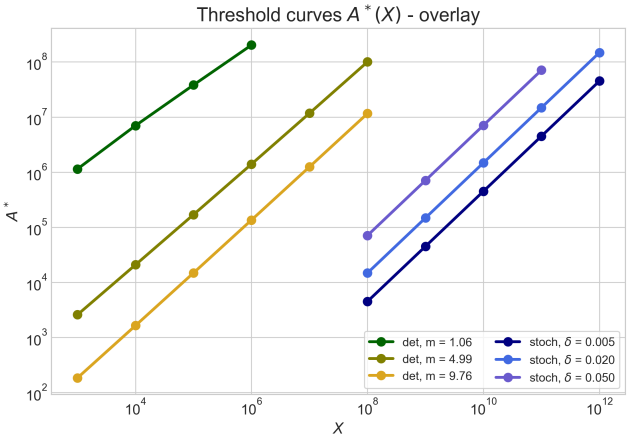
$$P(A^*(X)) = X$$

$A^*(X)$ is the **activity level required** for the token to match the outside asset.



Thresholds Across Models

3 When Does the Token Dominate?



WHAT THE OVERLAY SHOWS

- **Competitive pressure.** Higher X raises the benchmark the token must beat; at high X, the required activity increases disproportionately.
- **Effective horizon.** Longer maturity lowers the deterministic threshold, while higher termination risk raises the stochastic threshold.
- **Viability region.** When the threshold is too high, tokenization is not infeasible mathematically, but economically unattractive at realistic activity levels.



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A reserve-backed motivation inspired by RIVCoin

4 From Token Valuation to Token Design

Valuation So Far

Token **supply** is takes as **given**.

Price emerges **from equilibium** adoption and **market clearing**.

The analysis extended CWL along two missing dimensions:

- **Time horizon**
- **External competition**



Reserve-Backed Design

In reserve-backed systems, **issuance** is a **variable** policy.

Token **value** depends on

- a reserve-backed **intrinsic value**,
- an **activity-driven** valuation premium,
- an endogenous discount induced **issuance**.

The central **trade-off** is therefore: **growth today** vs. **dilution tomorrow**

Inspired by **RIVCoin**, Chapter 3 studies how issuance can be chosen to support value creation without weakening per-token backing.

Intrinsic Value and Issuance trade-off

4 From Token Valuation to Token Design

INTRINSIC VALUE

$$I_t = V_t / k_t$$

It represents the amount of **reserve backing** each token.



$$dV_t = \mu_V V_t\, dt + \sigma_V V_t\, dZ_t + P_t x_t k_t\, dt \qquad dk_t = x_t k_t\, dt$$

$$dI_t = [\mu_V I_t + x_t(P_t - I_t)]\, dt + \sigma_V I_t\, dZ_t$$

ISSUANCE

Protocol **decision lever**



x_t

New tokens issued by the protocol

TWO OPPOSITE EFFECTS

Reserve Accumulation

New issuance can attract resources and increase reserve value.

Dilution Pressure

Higher supply means more claims and can compress the premium.

MARKET PRICE

$$P_t = I_t \left(1 + \phi A_t^{\beta}\right) e^{-\gamma x_t}$$

- Anchored to **reserves**.
- Amplified by **activity**.
- Compressed by **issuance**.

Optimal Issuance Rate

4 From Token Valuation to Token Design

$$x_t^* = \arg\max_{x_t \in [0,\bar{x}]} J(x_t)$$

$$J(x_t) = \underbrace{\theta x_t(P_t - I_t) + \omega x_t k_t(P_t - I_t)}_{\text{Value creation}} - \underbrace{\chi(\Pi_t - 1)^2 - \frac{\lambda}{2}x_t^2}_{\text{Market discipline}}$$

Existence & Uniqueness

The admissible issuance set must satisfy:

$$\bar{x} \leq \frac{2}{\gamma}$$

$$\Pi_t(x_t) > \frac{1}{2}$$

$$\Downarrow$$

$$\forall x_t \in [0, \bar{x}]$$

Unique Optimal Issuance Rate

Closed-Form Approximation

For small or moderate values of γx_t:

$$e^{-\gamma x_t} \simeq 1 - \gamma x_t$$

$$\downarrow$$

First-order condition $\Rightarrow$ quadratic form

$$a_t x_t^2 + b_t x_t + c_t = 0$$

$$\downarrow$$

Discriminant

$$\Delta_t = b_t^2 - 4a_t c_t$$

$$\downarrow$$

When $\Delta_t \geq 0$, the relevant root is selected.

Closed-Form Issuance Rule

$$\hat{x}_t^* = \frac{-b_t - \sqrt{\Delta_t}}{2a_t}$$



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Key Takeaways

5 Conclusions

Time Horizon



Finite duration matters

External Competition



Token viability is relative, not absolute

Issuance Design



Supply policy turns valuation into design.

A token is not simply an asset to be priced.

It is an institutional architecture to be designed.

Next Step: Dynamic Token Pricing

5 Conclusions

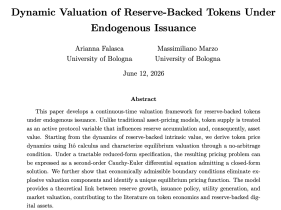
BEYOND THE THESIS

- Dynamic extension of the reserve-backed model.
- Intrinsic value I_t becomes the key state variable.
- Token price $P(I_t)$ is derived, not imposed.
- No-arbitrage pins down equilibrium valuation.
- Reserve growth and issuance jointly shape market value.

RESEARCH QUESTION

How does endogenous issuance reshape the dynamic valuation of reserve-backed tokens?

Working Paper



Dynamic Valuation of Reserve-Backed Tokens Under Endogenous Issuance

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June 12, 2026

Abstract

This paper develops a continuous-time valuation framework for reserve-backed tokens under endogenous issuance. Unlike traditional asset-pricing models, token supply is treated as an active protocol variable that influences reserve accumulation and, consequently, asset value. Starting from the dynamics of reserve-backed intrinsic value, we derive token price dynamics using Itô calculus and characterize equilibrium valuation through a no-arbitrage condition. Under a tractable reduced-form specification, the resulting pricing problem can be expressed as a second-order Cauchy-Euler differential equation admitting a closed-form solution. We further show that economically admissible boundary conditions eliminate explosive valuation components and identify a unique equilibrium pricing function. The model provides a theoretical link between reserve growth, issuance policy, utility generation, and market valuation, contributing to the literature on token economics and reserve-backed digital assets.

Keywords: reserve-backed tokens, tokenomics, digital assets, endogenous issuance, continuous-time finance, fintech.

The thesis establishes the architecture.
The paper develops its dynamic pricing implications.



Rethinking Token Valuation for Digital Platforms

Thank you!

Q&A